Morning Paper Report - Tuesday, October 20, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Comparative Evaluation of AI Models for Flood Prediction: CNN and RNN Integration with Bi-LSTM

Brijesh Munjiyasara, Dhvani Rana, Dhyey Patel, Smeet Patel, Yagnesh Vyas, Vijay Singh, et al.
International Journal of Scientific Research in Science and Technology, 2025

Source: Semantic Scholar; matched terms: modeling, neural network, neural networks, deep learning, ai model

Link: https://www.semanticscholar.org/paper/b86bab70a789176e47d8380734194ba1d8d80920

Goal: Flooding is devastating natural disaster that poses significant threats to human life, infrastructure, and economies, particularly in regions susceptible to heavy rainfall and monsoonal patterns

Method: The research evaluates modeling results for different forecasting horizons for Darbhanga, Bihar

Results: The research evaluates modeling results for different forecasting horizons for Darbhanga, Bihar

Conclusion: The research evaluates modeling results for different forecasting horizons for Darbhanga, Bihar The study draws on existing literature and proposes methodology to evaluate these models using ten years of spatial-temporal historical rainfall and flood data, aiming to provide valuable insights for governance, practitioners and researchers in field of flood disaster management.

2. AI Models Can Predict and Collaboratively Modulate Human Memory Search

E. Lacosse, Mariana Duarte, Graham Todd, Peter M. Todd, Daniel C. McNamee
2026

Source: Semantic Scholar; matched terms: artificial intelligence, ai model, human memory

Link: https://www.semanticscholar.org/paper/523ad627311d66b29e107c1f71f21bcb164e9c08

Goal: Large language models (LLMs) exhibit unprecedented natural language generation and many text-based problem-solving capabilities

Method: Large language models (LLMs) exhibit unprecedented natural language generation and many text-based problem-solving capabilities

Results: To test this, we use the semantic fluency task (SFT), a classic cognitive paradigm requiring generative semantic memory retrieval that has long served to characterize convergent and divergent thinking in humans We demonstrate that an LLM's abilities to track and predict human memory trajectories in this task exceed those of other humans.

Conclusion: To test this, we use the semantic fluency task (SFT), a classic cognitive paradigm requiring generative semantic memory retrieval that has long served to characterize convergent and divergent thinking in humans We demonstrate that an LLM's abilities to track and predict human memory trajectories in this task exceed those of other humans.

3. Modeling Working Memory Capacity in Sentence Processing with Lossy-Context Surprisal

Jiaxin Li, Michael Hahn, Vera Demberg
2026

Source: Crossref; matched terms: working memory, computational model, modeling

Link: https://doi.org/10.31234/osf.io/u5tf2_v1

Goal: Understanding how comprehenders interpret linguistic input in real time has long been a focus of sentence processing research

Method: The recently-proposed Resource-Rational Lossy-context Surprisal (RR-LCS) model can be applied for making predictions at the level of individual readers: it predicts that individuals with low working memory capacity (WMC) have a lossy representation of the previous context and may therefore experience high surprisal at a long-distance dependency, while individuals with high WMC will experience a lower surprisal...

Results: The results revealed that low-WMC participants initially showed a strong locality effect between the short and long conditions, which decreased over time due to adaptation

Conclusion: The recently-proposed Resource-Rational Lossy-context Surprisal (RR-LCS) model can be applied for making predictions at the level of individual readers: it predicts that individuals with low working memory capacity (WMC) have a lossy representation of the previous context and may therefore experience high surprisal at a long-distance dependency, while individuals with high WMC will experience a lower surprisal...

4. Visuospatial Working Memory for Autonomous UAVs: A Bio-Inspired Computational Model

José-Antonio Cervantes, Sonia López, Salvador Cervantes, Adriana Mexicano, Jonathan-Hernando Rosales
Applied Sciences, 2021

Source: Crossref; matched terms: working memory, computational model

Link: https://doi.org/10.3390/app11146619

Goal: Visuospatial working memory is a fundamental cognitive capability of human beings needed for exploring the visual environment

Method: A bio-inspired computational model of Visuospatial Working Memory (VSWM) is proposed in this paper to endow Autonomous Unmanned Aerial Vehicles (UAVs) with this cognitive function

Results: The VSWM model was capable of maintaining a coherent and continuous representation of visual and spatial relationships among interest objects presented in the environment even when a visual stimulus is lost because of a total occlusion The VSWM model proposed in this paper represents a step towards autonomous UAVs capable of forming visuospatial mental imagery in realistic environments.

Conclusion: The VSWM model was capable of maintaining a coherent and continuous representation of visual and spatial relationships among interest objects presented in the environment even when a visual stimulus is lost because of a total occlusion The VSWM model proposed in this paper represents a step towards autonomous UAVs capable of forming visuospatial mental imagery in realistic environments.

5. Modeling Individual Differences in Working Memory: Subject-Level Parameter Recovery within the Memory Measurement Model Framework (M^3)

Jan Göttmann, Gidon T. Frischkorn, Klaus Oberauer, Simon B. Schaefer, Anna-Lena Schubert
2025

Source: Crossref; matched terms: working memory, precision, modeling

Link: https://doi.org/10.31234/osf.io/945d2_v1

Goal: The memory measurement model (M3) is a flexible modeling framework designed to measureparameters associated with different processes in working memory

Method: The memory measurement model (M3) is a flexible modeling framework designed to measureparameters associated with different processes in working memory

Results: Based on theresults, we recommend that researchers should ensure that participants select not-presented lures(i.e., items not in the memory set) at least sometimes (≥ 2%) and collect a minimum of 200retrieval responses per subject (40 trials at set size 5)

Conclusion: In contrast, M3 parameters modelingfree-time effects such as extended encoding and removal processes could not be recovered withsufficient precision and showed miscalibrated posteriors and poor reliability These challengesprimarily arose from the substantial amount of data required to achieve sufficiently preciseparameter estimates.